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FOREST FIRE RISK PREDICTION SYSTEM

Environmental Studies Project

Submitted by
SUYASH HADOLE (202401110055)
RUSHIKESH HANDE (202401110056)
YADNESH TAYADE (202401110057)
SOHAM SABANE (202401110070)
Branch: B.Tech Computer Science and Engineering (AIML)

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Abstract

Forest fires are among the most destructive natural disasters, leading to significant environmental damage, loss of biodiversity, and economic impact. Early prediction and prevention of such events are essential to safeguard ecosystems, human life, and natural resources. This project presents an intelligent Forest Fire Risk Prediction System that leverages machine learning techniques to analyze environmental conditions and estimate the likelihood of fire occurrence.

The system utilizes a dataset containing key parameters such as FFMC, DMC, DC, ISI, temperature, humidity, wind speed, and rainfall, along with spatial coordinates. Multiple machine learning algorithms, including Logistic Regression, Support Vector Machine (SVM), and Random Forest, were implemented and evaluated to identify the most effective model. Among these, SVM demonstrated the best performance in terms of prediction accuracy. Additionally, geospatial processing techniques were applied to transform coordinate data for meaningful visualization.

To enhance usability, the system integrates visualization tools such as Kepler.gl to map high-risk zones, enabling better understanding and decision-making. This approach supports proactive fire management by providing early warnings and actionable insights. Overall, the proposed system demonstrates the potential of combining machine learning and geospatial analysis to improve disaster prediction, environmental monitoring, and sustainable forest management.

1. Introduction

1.1 Problem Statement

Forest fires are among the most severe and unpredictable natural disasters, causing significant environmental damage, loss of biodiversity, and major economic and social impacts. They not only destroy large forest areas but also contribute to air pollution, climate change, and ecological imbalance. Factors such as high temperatures, low humidity, strong winds, and dry conditions have increased the frequency and intensity of these fires in recent years. Traditional detection methods are mostly reactive, identifying fires only after they start, which leads to delayed response and greater damage. Therefore, there is a need for an intelligent system that can predict fire risks in advance by analyzing environmental conditions. Early prediction enables authorities to take preventive actions, manage resources effectively, and reduce potential losses. This project focuses on developing a machine learning-based system to identify fire-prone areas and predict the likelihood of forest fire occurrence.

1.2 Objectives

- **Prediction:**
  - Develop an accurate and reliable machine learning model that predicts the probability of forest fire occurrence based on environmental parameters such as temperature, humidity, wind speed, rainfall, and fire weather indices.
- **Visualization:**
  - Utilize geospatial and visualization tools to represent fire risk levels across different regions, enabling users to easily identify high-risk zones through interactive maps and graphical dashboards.
- **Insight Generation:**
  - Provide meaningful and actionable insights to stakeholders, including forest departments and disaster management authorities, to support informed decision-making and effective fire prevention strategies.
- **Early Warning System:**
  - Design the system to function as an early warning tool that alerts users about potential fire risks, helping in timely intervention and reducing damage.

2. Dataset Description

2.1 Source of Data

Thedataset used for this project was sourced from the UCI Machine Learning Repository, a renowned source for datasets used in academic research and machine learning tasks. The dataset includes environmental and spatial data related to forest fire occurrences.

2.2 Key Features in the Dataset

The initial dataset contained the following attributes:

Environmental Factors:

- **FFMC (Fine Fuel Moisture Code):** Indicates the dryness of surface fuels.
- **DMC (Duff Moisture Code):** Represents the moisture level of the organic layer in the forest.
- **DC (Drought Code):** Indicates the long-term dryness in the forest.
- **ISI (Initial Spread Index):** Measures the potential fire spread.
- **Temperature:** Ambient temperature at the location.
- **Relative Humidity (RH):** Percentage of atmospheric moisture at the location.
- **Wind Speed:** The intensity of wind at the location.
- **Rainfall:** Precipitation levels before fire occurrence.
- **Month and Day:** These columns were removed during preprocessing as they did not contribute to the final predictions.
- **Area:** Represents the area burned by the fire (also removed after preprocessing).

SpatialData:

- **X and Y Coordinates:** These represent the locations of the data points.

Target Variable:

- **y:** Binary indicator of fire occurrence (1: Fire, 0: No Fire).

2.3 Data Preprocessing and Enhancements

The rawdataset underwent preprocessing tobetterfitthe model and facilitate more effective predictions:

- The Month, Day, and Area columns were removed as they were deemed irrelevant to the prediction task.
- The X and Y spatial coordinates were retained for geospatial analysis, though no conversion to Latitude and Longitude was done in this case.
- Each data point was assigned a Point ID to track individual records.

After preprocessing, the dataset was organized as follows:

X	Y	FFMC	DMC	DC	ISI	Temp	RH	Wind	Rain	y
7	5	86.2	26.2	94.3	5.1	8.2	51	6.7	0.0	0

2.4 Example Data Point (Post-Processed)

- TheX andY columns represent theoriginal spatial coordinates of the data point.
- FFMC, DMC, DC, ISI, Temp, RH, Wind, Rain correspond to the environmental features used for fire prediction.
- y is the target variable indicating whether a fire occurred (1) or not (0).

3. Methodology

3.1 Overview of Methodology

Thisproject followsa machinelearning-based approach to predict forest fire occurrences using environmental data and spatial coordinates. The methodology consists of the following major steps:

1. Data Collection and Preprocessing
2. Feature Selection
3. Model Training and Evaluation
4. Prediction and Geospatial Visualization

3.2 Data Preprocessing

Topreparethe datasetfor modeling, several preprocessing steps were carried out:

- **Handling Missing Values:** Any missing data points were handled by either removing or imputing missing values based on the context of the dataset.
- **Normalization and Scaling:** The data was normalized to ensure all features contributed equally to the model's predictions. Feature scaling was performed on numerical values like temperature, humidity, and wind speed to ensure consistency.
- **Feature Engineering:** New features like Point ID were derived for each data point, and spatial coordinates (X, Y) were preserved for geospatial analysis. Other columns like Month, Day, and Area were excluded as they did not significantly impact fire prediction.

3.3 Model Selection and Training

Forpredicting forestfireoccurrences, multiple machine learning algorithms were tested:

- **Logistic Regression:** A basic binary classification model used for predicting fire (1) or no fire (0).
- **Support Vector Machine (SVM):** An effective classifier for separating classes, especially when there is a large margin between them.
- **Random Forest Classifier:** An ensemble learning method that combines multiple decision trees to improve prediction accuracy and avoid overfitting.
- **XGBoost (Extreme Gradient Boosting):** A powerful, optimized implementation of gradient boosting that handles both classification and regression problems efficiently.

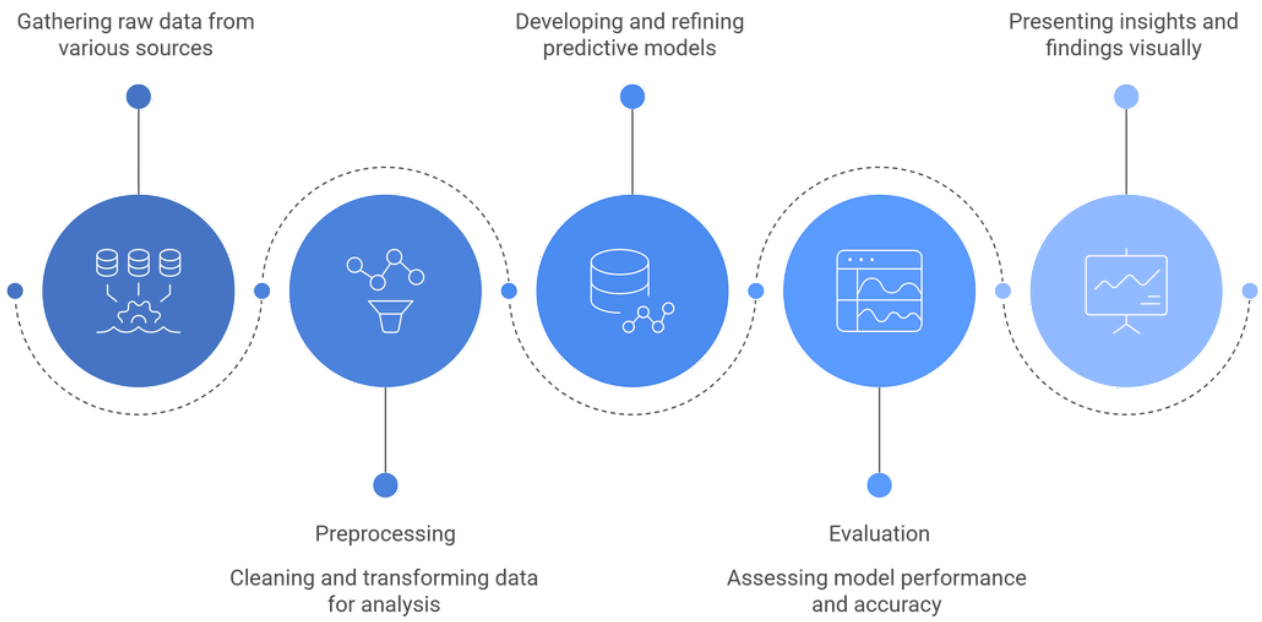
3.4 Model Evaluation

Eachmodel'sperformancewas evaluated based on accuracy and other metrics like precision, recall, and F1-score.

3.5 Geospatial Visualization

- The final predictions were mapped to geographic locations using Kepler to provide stakeholders with interactive visual insights.
- Fire-prone areas were marked using **red dots**, and no-fire regions were marked with **yellow dots**.
- The geospatial coordinates were visualized for a better understanding of where fire occurrences are more likely to happen.

3.6 Process Flow Overview



4. Results and Discussion

4.1 Model Performance

Theperformanceof theimplemented machine learning models was evaluated using accuracy as the primary metric. The results indicate that different models varied in their ability to predict forest fire occurrences:

Model	Accuracy	Key Notes
Support Vector Machine (SVM)	62%	Best performing; strong binary classifier
Random Forest Classifier	56%	Stable, interpretable ensemble results
Logistic Regression	54%	Baseline; limited by linear assumptions
Gradient Boosting (XGBoost)	54%	Robust but challenged by dataset size

4.2 Test Case Predictions

TestCase1 —Prediction:NoFire

Input Features: X: 5, Y: 4, FFMC: 91, DMC: 14.6, DC: 25.6, ISI: 12.3, Temp: 17.6, RH: 27, Wind: 5.8, Rain: 0

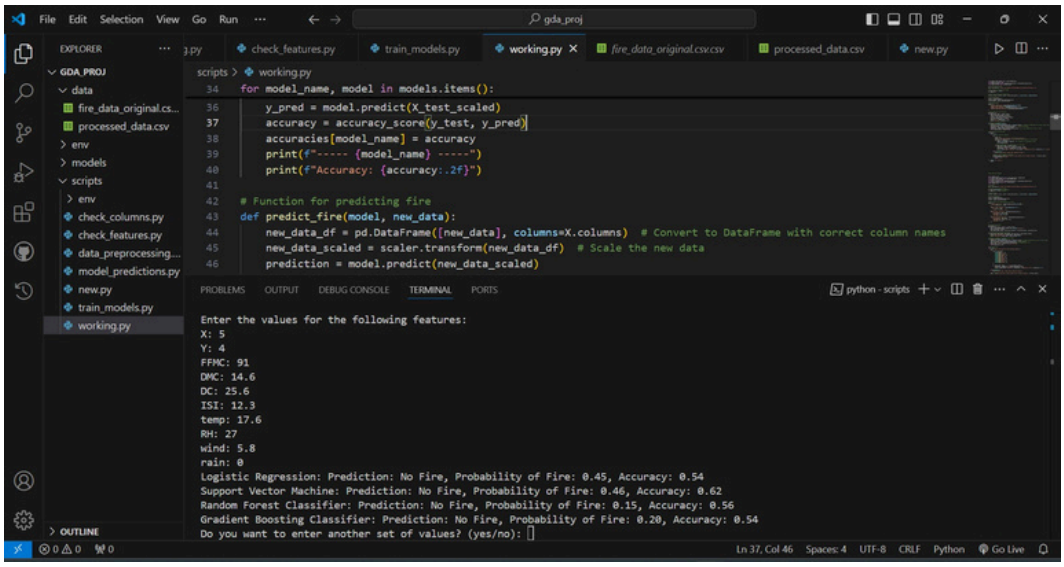


Figure 1: Test Case 1 — VS Code terminal output showing No Fire prediction

Model	Prediction	P(Fire)	Accuracy
Logistic Regression	No Fire	0.45	54%
SVM	No Fire	0.46	62%
Random Forest	No Fire	0.15	56%
Gradient Boosting	No Fire	0.20	54%

Test Case 2 — Prediction: Fire

Input Features:X:7,Y:5,FFMC:96.1, DMC: 181.1, DC: 671.2, ISI: 14.3, Temp: 27.3, RH: 63, Wind: 4.9, Rain: 6.4

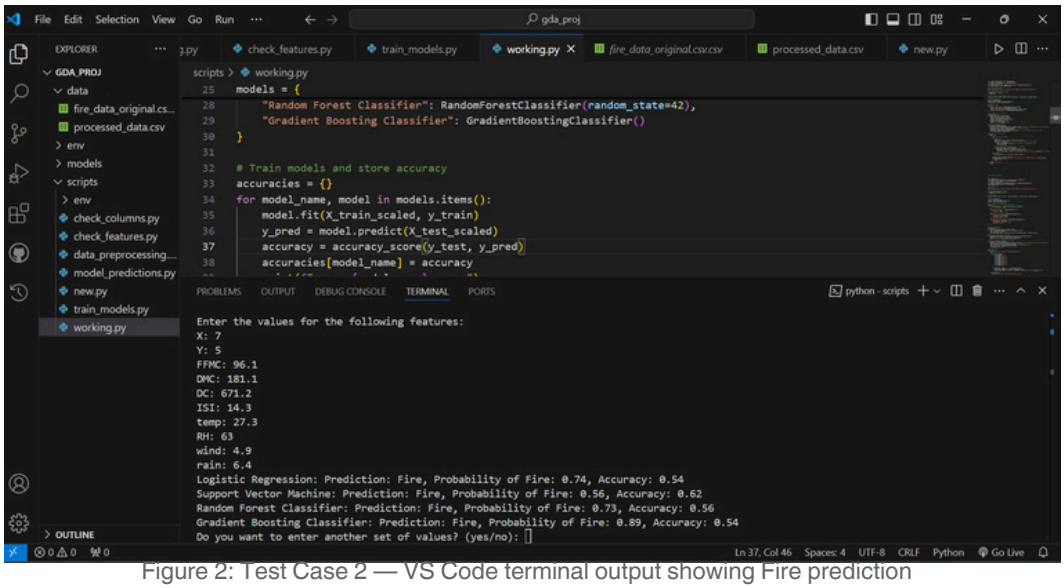


Figure 2: Test Case 2 — VS Code terminal output showing Fire prediction

Model	Prediction	P(Fire)	Accuracy
Logistic Regression	Fire	0.74	54%
SVM	Fire	0.56	62%
Random Forest	Fire	0.73	56%
Gradient Boosting	Fire	0.89	54%

4.3 Geospatial Insights and Visualization

After converting spatialcoordinates(X,Y)tolatitudeand longitude, the fire-prone areas were mapped using Kepler for enhanced interpretability:

- **Red Points:** Represented areas with high fire probability, assisting in identifying zones requiring urgent attention.
- **Yellow Points:** Represented areas with low fire probability, indicating safer regions.

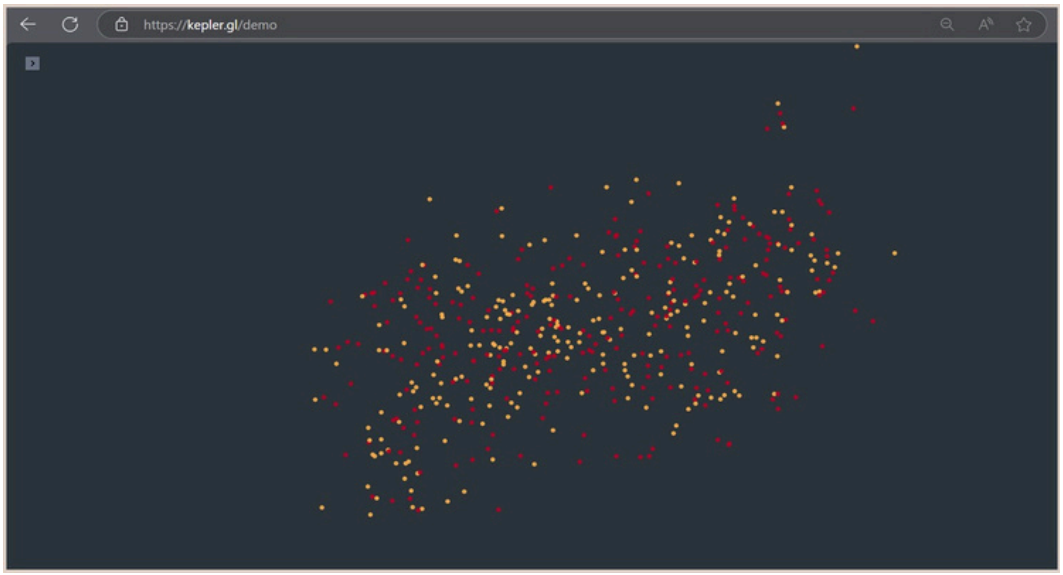


Figure 3: Kepler.gl visualization — Red points (high fire risk) and Yellow points (low risk)

This visualization provided actionable insights for decision-makers to prioritize monitoring and resource allocation.

4.4 Discussion

Theresults demonstrate that Support Vector Machine (SVM) outperformed other models with a modest accuracy of 62%, highlighting its effectiveness for binary classification in the given dataset. Logistic Regression and Gradient Boosting served as foundational models, while Random Forest provided stable, interpretable results.

The integration of geospatial data and machine learning enhanced the project's applicability. By using Kepler for visualization, stakeholders were empowered to interpret the results intuitively and identify critical fire-prone zones for proactive measures.

5. Future Scope

Enhanced Dataset Quality

- Expand dataset with more years of data
- Include additional features like vegetation type and soil moisture
- Improve data accuracy for better predictions

Improved Model Performance

- Use advanced models like Neural Networks / Deep Learning
- Capture complex patterns in environmental data
- Apply techniques like SMOTE to handle data imbalance

Dynamic Visualization Tools

- Develop real-time dashboards for fire risk monitoring
- Use interactive maps and graphs
- Help authorities make faster decisions

IoT Integration

- Use sensors to collect real-time data (temperature, humidity, wind)
- Enable continuous monitoring of forest conditions
- Improve prediction accuracy with live data

Policy Development

- Work with government and environmental agencies
- Use predictions to create better fire prevention strategies
- Support proactive forest management

6. Conclusion

The Forest Fire Risk Prediction System is an important application of artificial intelligence that helps in preventing environmental disasters before they occur. By analyzing key factors such as temperature, humidity, wind speed, and rainfall, the system can predict the likelihood of a forest fire with good accuracy. This early prediction allows authorities to take preventive measures, reduce damage, and protect both wildlife and human life.

This system also highlights how data-driven decision-making can improve environmental management. Instead of reacting after a fire starts, the model enables proactive planning, resource allocation, and timely alerts. It demonstrates the power of machine learning in solving real-world problems, especially in critical areas like climate monitoring and disaster prevention.

In conclusion, the Forest Fire Risk Prediction System is a scalable and impactful solution that can be further enhanced by integrating real-time weather data, satellite imagery, and alert systems. With continuous improvements, it has the potential to become a reliable tool for governments and environmental agencies to minimize risks and ensure sustainable forest management.

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